

Def Let $D \in \mathbb{Z}$ be a square-free integer.

Then, a set $\mathbb{Q}(\sqrt{D}) = \{a + b\sqrt{D} \mid a, b \in \mathbb{Q}\}$ forms a field, and

$\mathbb{Z}[\sqrt{D}] = \{a + b\sqrt{D} \mid a, b \in \mathbb{Z}\}$ forms a ring, as a subring of $\mathbb{Q}(\sqrt{D})$.

Further, if $D \equiv 1 \pmod{4}$, then for $\omega = \frac{1 + \sqrt{D}}{2}$,

$$\mathbb{Z}[\omega] = \{a + b\omega \mid a, b \in \mathbb{Z}\}$$

is also subring of $\mathbb{Q}(\sqrt{D})$, and contains $\mathbb{Z}[\sqrt{D}]$. Figure out now.

At first, $\mathbb{Q}(\sqrt{D})$ is vector space over $\mathbb{Q}$, with basis $\{1, \sqrt{D}\}$. So, define a Norm as

$$N: \mathbb{Q}(\sqrt{D}) \rightarrow \mathbb{Q}; a + b\sqrt{D} \mapsto a^2 - D \cdot b^2$$

Actually, $N(\alpha) = \prod_{\sigma \in \text{Gal}(\mathbb{Q}(\sqrt{D})/\mathbb{Q})} \sigma(\alpha)$ So, if $\alpha \neq 0$, $\alpha^{-1} = \frac{\overline{\alpha}}{N(\alpha)}$.

In $\mathbb{Q}(\sqrt{D})$, It is clearly abelian group under the addition, and by the fact above, $\mathbb{Q}(\sqrt{D}) - \{0\}$ is abelian group under the multiplication.

Now, let's dissect $\mathbb{Z}[\sqrt{D}]$. Clearly $\mathbb{Z}[\sqrt{D}]$ is closed under the subtraction.

And, for $a + b\sqrt{D}$, $c + d\sqrt{D} \in \mathbb{Z}[\sqrt{D}]$,

$$(a + b\sqrt{D}) - (c + d\sqrt{D}) = (a - c) + (b - d)\sqrt{D} \in \mathbb{Z}[\sqrt{D}], \text{ so closed under the subtraction.}$$

Associative, Commutative, Containing 0 and 1 are clear.

Moreover, for $\mathbb{Z}[\omega]$, it is clearly group under the addition, and,

$$\left(a + b \cdot \frac{1 + \sqrt{D}}{2}\right) - \left(c + d \cdot \frac{1 + \sqrt{D}}{2}\right) = (a - c) + (b - d) \cdot \frac{D - 1}{4} + (a + b + c + d) \cdot \frac{1 + \sqrt{D}}{2}$$

is contained in $\mathbb{Z}[\omega]$, since $(D - 1)/4 \in \mathbb{Z}$, by $D \equiv 1 \pmod{4}$.

(More observation: for $a + b\sqrt{D}$ with $b \neq 0$, a minimal polynomial over the given ring is

$$(x - a)^2 - D \cdot b^2 = x^2 - 2ax + a^2 - D \cdot b^2$$

which has another root as $a - b\sqrt{D}$.)

Thm. The field Norm $N: \mathbb{Q}(\sqrt{D}) \rightarrow \mathbb{Q}; a+b\sqrt{D} \mapsto a^2 - Db^2$ preserves the multiplication.
 Proof. Let $a+b\sqrt{D}, c+d\sqrt{D} \in \mathbb{Q}(\sqrt{D})$ be given. Then,

$$\begin{aligned} N[(a+b\sqrt{D}) \cdot (c+d\sqrt{D})] &= N(ac+bdD+(ad+bc)\sqrt{D}) \\ &= (ac+bdD)^2 - D \cdot (ad+bc)^2 \\ &= a^2c^2 + b^2d^2D^2 + 2abcd \cdot D - D \cdot (a^2d^2 + b^2c^2 + 2abcd) \\ &= a^2c^2 + b^2d^2D^2 - D \cdot (a^2d^2 + b^2c^2) \\ &= (a^2 - Db^2) \cdot (c^2 - Dd^2) \\ &= N(a+b\sqrt{D}) \cdot N(c+d\sqrt{D}). \end{aligned}$$

Thm. In $\mathbb{Z}[\sqrt{D}]$ or $\mathbb{Z}[u]$, an element α is a unit iff $N(\alpha) = \pm 1$.

Proof. If α is unit, then $N(\alpha \cdot \alpha^{-1}) = N(\alpha) \cdot N(\alpha^{-1}) = 1$.

But, since $N(\alpha), N(\alpha^{-1}) \in \mathbb{Z}$, both are 1 or both are -1.

Conversely, if $N(\alpha) = \pm 1$, $\frac{\alpha}{N(\alpha)}$ is inverse of α .